How this book is organized: A road map

This book has 11 chapters. Chapter 1, “Introduction to agents and their world,” begins by laying a foundation with fundamental definitions of large language models, chat systems, assistants, and autonomous agents. As the book progresses, the discussion shifts to the key components that make up an agent and how these components work together to create truly effective systems. Here is a quick summary of chapters 2 through 11:

- *Chapter 2, “Harnessing the power of large language models”*—We start by exploring how to use commercial LLMs, such as OpenAI. We then examine tools, such as LM Studio, that provide the infrastructure and support for running various open source LLMs, enabling anyone to experiment and innovate.
- *Chapter 3, “Engaging GPT assistants”* —This chapter dives into the capabilities of the GPT Assistants platform from OpenAI. Assistants are foundational agent types, and we explore how to create practical and diverse assistants, from culinary helpers to intern data scientists and even a book learning assistant.
- *Chapter 4, “Exploring multi-agent systems”* —Agentic tools have advanced significantly quickly. Here, we explore two sophisticated multi-agent systems: CrewAI and AutoGen. We demonstrate AutoGen’s ability to develop code autonomously and see how CrewAI can bring together a group of joke researchers to create humor collaboratively.
- *Chapter 5, “Empowering agents with actions”* —Actions are fundamental to any agentic system. This chapter discusses how agents can use tools and functions to execute actions, ranging from database and application programming interface (API) queries to generating images. We focus on enabling agents to take meaningful actions autonomously.
- *Chapter 6, “Building autonomous assistants”* —We explore the behavior tree—a staple in robotics and game systems—as a mechanism to orchestrate multiple coordinated agents. We’ll use